Yaming Zhang

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EDUCATION

Northeastern University, *Boston, MA* *Dec 2023*
Master of Science in Computer Engineering
Concentration: Computer Systems and Software

University of California Riverside, *Riverside, CA* *Jun 2020*
Graduate Preparation Program in Computer Science

Xidian University, *Xi'an, China* *Jun 2019*
Bachelor of Science in Computer Science

SKILLS

Languages: Java, Go, Node.js, Scala
Frameworks: Spring, gRPC, React, Kafka, Flink, Spark, EMR
Databases: MySql, Redis, MongoDB, Elasticsearch, ScyllaDB
DevOps: AWS, Kubernetes, ELK Stack, Prometheus, Grafana

WORK EXPERIENCE

Electronic Arts (EA), *Redwood City, CA*
Full-Stack Software Engineer Intern *May 2023 – Aug 2023*

- Designed the **Java-based OOP** model for Player Lifecycle Graph, a serialization model to manage players' states and unified messages, resulting in a **30% cost reduction** by eliminating the need for customizing **Braze**.
- Developed a **Flink** topology for **Kafka** streaming PIN events execution, enabling efficient retrieval of lifecycles from **Redis**, reading player profiles from **ScyllaDB**, and executing the corresponding lifecycles.
- Leveraged EADP's **Data Query Language (DQL)** to extract data from unstructured **JSON** events, evaluate boolean conditions, and execute action scripts.
- Built a web application with **React** and implemented APIs with **Spring Boot**, empowering campaign managers to draw lifecycle flowcharts and retrieve players' current states in games.

Tencent, *Beijing, China*
Software Engineer *Jul 2020 – Aug 2022*

- Developed **Microservices** on **Go** using **gRPC** with logs on **ELK**, monitoring on **Prometheus**, asynchronous pipelines to **Kafka**, etc, achieving a **25% improvement** in interface response time and a **50% reduction** in production debug time.
- Established automated **CI/CD** pipelines that enhanced service security by seamlessly integrating **Unit Tests** during CI, while also enabling efficient rollback on **Tencent Cloud Kubernetes**.
- Wrote customized codes in the secondary development of **Superset** to create a dashboard displaying user usage data of Tencent services by preprocessing and visualizing data with **Pandas** and **ECharts**.
- Analyzed the traffic data of Tencent news using **Scala Spark** and **PySpark** to promote automatic advertising and personalized recommendation.

The Big-Data Lab at UCR, *Riverside, CA*
Research Assistant & Scala Developer *Aug 2019 – Jul 2020*

- Leveraged the power of **Scala** and **Spark** to design user-defined functions for geometry calculation, enabling the provision of diverse and flexible APIs that empower developers to generate and manipulate geometries with ease.
- Utilized the **Hadoop** file API to efficiently parse records from the **OpenStreetMap** PBF file, enabling extraction of OSM entities for subsequent **Big Data Processing Pipelines** on **AWS EMR**.
- Extracted nodes, POIs, roads and all objects in pipelines and categorized different datasets from all objects based on OSM tags, including parks, lakes and buildings, etc.
- Generated the satellite image of corresponding OSM Vector data using **Beast Raptor** spatial-join for spatial machine learning and AI activities.

PROJECT

High Performance Computing *Sept 2022 – Dec 2022*

- Developed **C++** code employing **OpenMP**, **OpenMPI**, and **CUDA** for matrix multiplication, achieving superior performance via parallel computing strategies.
- Utilized diverse languages: **C++**, **Python**, **Go**, **Julia** for vector addition and matrix multiplication, benchmarking multithreading, **GPU** acceleration, and **BLAS/GEMM** functions.
- Evaluated and compared code performance on varying **AMD** and **Intel** CPUs, offering insights into execution speed disparities and resource utilization.

PUBLICATIONS

Samriddhi Singla, **Yaming Zhang**, Ahmed Eldawy (2022). OSMX: Spark-based Geospatial Data Extractor from OpenStreetMap and Satellite Images. International Conference on Advances in Geographic Information Systems.

Ahmed Eldawy, ..., **Yaming Zhang** (2021). Beast: Scalable Exploratory Analytics on Spatio-temporal Data. ACM International Conference on Information and Knowledge Management.

Yaming Zhang, Ahmed Eldawy (2020). Evaluating Computational Geometry Libraries for Big Spatial Data Exploration. International ACM SIGMOD Workshop on Managing and Mining Enriched Geo-Spatial Data.